

E - Fibonacci String

分值: 450 分

Problem Statement

You are given strings X and Y . Define a sequence of strings $S_1, S_2, \dots$ as follows.

给定字符串 X 和 Y 。按如下方式定义字符串序列 $S_1, S_2, \dots$:

- $S_1 = X$
- $S_2 = Y$
- For $i \geq 3$, S_i is the concatenation of S_{i-1} and S_{i-2} in this order.
- $S_1 = X$
- $S_2 = Y$
- 对于 $i \geq 3$, S_i 是 S_{i-1} 和 S_{i-2} 按此顺序拼接得到的结果。

For each $i = 1, 2, \dots, Q$, answer the following problem.

对于每个 $i = 1, 2, \dots, Q$, 回答以下问题。

Problem: You are given integers L_i, R_i and a character C_i . Find how many times character C_i appears in the L_i -th through R_i -th characters of $S_{10^{18}}$.

问题: 给定整数 L_i, R_i 和一个字符 C_i 。求 $S_{10^{18}}$ 的第 L_i 到第 R_i 个字符中, 字符 C_i 出现的次数。

Constraints

- X and Y are strings of lowercase English letters of length between 1 and 10^4 , inclusive.
- $1 \leq Q \leq 10^5$
- $1 \leq L_i \leq R_i \leq 10^{18}$
- C_i is a lowercase English letter.
- All given numerical values are integers.

- X 和 Y 是由小写英文字母组成的字符串，长度在 1 到 10^4 之间（含两端）。
 - $1 \leq Q \leq 10^5$
 - $1 \leq L_i \leq R_i \leq 10^{18}$
 - C_i 是小写英文字母。
 - 所有给出的数值均为整数。
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Input

The input is given from Standard Input in the following format:

输入以如下格式从标准输入给出：

```
X
Y
Q
L1 R1 C1
L2 R2 C2
⋮
LQ RQ CQ
```

Output

Output Q lines.

The i -th line should contain how many times character C_i appears in the L_i -th through R_i -th characters of $S_{10^{18}}$.

输出 Q 行。第 i 行应包含 $S_{10^{18}}$ 的第 L_i 到第 R_i 个字符中，字符 C_i 出现的次数。

Sample Input 1

```
a
b
6
2 7 a
1 3 b
3 7 b
1 9 c
```

```
1 10000000000000000000 b
10000000000000000000 10000000000000000000 a
```

Sample Output 1

```
3
2
3
0
618033988749894848
1
```

S_3, S_4, S_5 are ba, bab, babba, respectively.

S_3, S_4, S_5 分别为 ba、bab、babba。

$S_{10^{18}}$ is babbababbabba..., and the second through seventh characters contain three occurrences of a.

$S_{10^{18}}$ 为 babbababbabba..., 其第 2 到第 7 个字符中包含 3 个 a。